

ResumeAI

AI Resume Builder + AI Resume Coach

Product Requirements Document • Version 1.0 • 2025

Next.js

Node.js

MySQL

OpenAI API

Tailwind CSS

Indian Market First • Hinglish-Friendly UX • ATS-Optimized

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1. Product Overview & Vision

What we're building, why it matters, and who we're building it for

Vision: Ek aisa AI-powered platform jo har Indian student aur professional ko recruiter-ready resume banane mein help kare — bina career counselor ke.

1.1 Mission Statement

ResumeAI ka mission hai har job seeker ko — chahe fresher ho, career switcher ho, ya non-English speaker — ek intelligent AI coach dena jo unka resume real-time mein improve kare, ATS systems se paar karaye, aur unhe interview tak pohchaye.

1.2 Problem Statement

India mein har saal 1.5 crore+ graduates job market mein aate hain. In mein se 78% ke paas professionally written resume nahi hota. Current resume builders: templates dete hain lekin guidance nahi. ATS rejection rate 75% se zyada hai. Professional resume writers expensive hain (₹2,000–₹15,000). Non-English speakers ke liye aur bhi mushkil hai.

Problem	Current Reality
Wrong Tool	Canva/Word/Google Docs — design tools, not career tools
No AI Guidance	Existing builders fill forms, not improve content
ATS Blindness	Beautiful resumes that get auto-rejected by ATS bots
Generic Output	Same template for a fresher and a 10-year veteran
Language Barrier	English-only UX alienates 60% of Indian job seekers
No Feedback Loop	Users submit resumes without knowing why they fail

1.3 Unique Value Proposition

AI Coach

Real-time bullet improvement & weak phrase detection

ATS Engine

Live ATS score + JD keyword matching before you export

Hinglish UX

India-first interface with bilingual coaching messages

1.4 Target Audience

Segment	Profile	Context
Primary	College students & fresh graduates (18–24)	India — Tier 1, 2, 3 cities
Secondary	Early professionals (1–5 years exp)	IT, Finance, Marketing
Tertiary	Career switchers & MBA students	Upskilling segment

Segment	Profile	Context
Special	Non-English speakers, regional language users	Hindi belt, South India
B2B	Colleges, bootcamps, placement cells	Institution partnerships

2. Market Research & Competitor Analysis

Market Size: India's online career services market is █4,200 Cr (2024) growing at 18% CAGR. Resume builder TAM globally: \$1.2B by 2027.

2.1 Competitor Analysis

Competitor	Strengths	Weaknesses
Canva	Beautiful templates, brand recognition	No AI coaching, not ATS-friendly, design over substance
Zety	Strong ATS tips, good content suggestions	Expensive (\$23/mo), no Indian market focus, English only
Rezi	Best ATS optimization, keyword analysis	Complex UX, no real-time AI coach, expensive for students
Kickresume	AI writing, good templates, LinkedIn import	Weak ATS scoring, no JD matching, limited free tier
Resume.io	Clean UX, fast creation	No AI coach, generic suggestions, no Indian context
Naukri Resume	India-specific, free	Outdated UI, no AI features, poor PDF quality

2.2 Market Gap — Our Opportunity

- **No Indian-first AI resume coach:** existing tools ignore Hinglish UX, Indian job portals (Naukri, LinkedIn India), and Indian resume conventions
- **Real-time coaching gap:** all competitors are form-fill tools — none provide inline AI suggestions as you type
- **ATS + JD matching together:** no tool combines live ATS scoring WITH specific JD keyword extraction in one flow
- **Affordability:** █0–█499/month vs \$15–\$25/month — massive price gap for students
- **Resume Roast / Scoring:** gamified feedback mechanism that no competitor offers

3. User Personas

Five detailed personas driving every product decision

■ Priya — College Fresher

Age / Location	21, Jaipur, B.Tech CSE final year
Goal	First job in IT — Infosys, TCS, or startup
Pain Points	Never written a resume; doesn't know what ATS is; writes 'I am hardworking' in every bullet
Tech Comfort	High with mobile, medium with web apps
Needs from ResumeAI	Step-by-step guidance, AI questioning ('Tell me your best project'), Hinglish tips
Success Metric	Lands first interview within 30 days of using platform

■ Rahul — Software Engineer (3 yrs)

Age / Location	25, Bengaluru, working at mid-size IT company
Goal	Switch to product company — Flipkart, Swiggy, or FAANG
Pain Points	Has experience but can't quantify achievements; resume reads like a job description
Tech Comfort	Very high
Needs from ResumeAI	Bullet rewriter, impact quantification, ATS score for specific JDs
Success Metric	Resume score improves from 54 → 85+ for target JD

■ Meena — Career Switcher

Age / Location	29, Pune, moving from banking to product management
Goal	Reposition 5 years of banking experience for PM roles
Pain Points	Transferable skills not highlighted; resume looks wrong for new field
Tech Comfort	Medium
Needs from ResumeAI	JD-based tailoring, skill gap analysis, AI rewrite for sector change
Success Metric	Gets PM interview callbacks within 60 days

■ Arjun — Non-English Speaker

Age / Location	23, Patna, Hindi medium background
Goal	Government / PSU job or private sector entry-level
Pain Points	Struggles with professional English; grammar errors; doesn't know industry keywords
Tech Comfort	Medium-low
Needs from ResumeAI	Hinglish UI, grammar correction, simple English suggestions, basic templates
Success Metric	Creates grammatically correct, professional resume without external help

■ Shreya — Recruiter (Platform Insight Persona)

Role	Senior HR, 200-person tech startup, receives 300 resumes/week
Frustration	80% resumes are generic; no quantified achievements; ATS spammed with keyword stuffing
What she wants to see	Impact numbers, clean formatting, role-relevant skills, honest profiles
Value for platform	Her perspective shapes our AI scoring rubric and ATS rules engine

4. Core Features — Deep Dive

Every feature explained with implementation detail

4.1 AI Resume Builder

- AI-guided section-by-section creation flow (not a blank form)
- AI Questioning System: asks user about experience, projects, achievements via chat interface
- Extracts structured data from conversational answers using GPT function calling
- Auto-fills resume sections from user's chat responses
- Smart defaults based on role/industry/experience level selection

4.2 Real-Time AI Suggestions

- Inline suggestions appear as user types (500ms debounce)
- Weak phrase detection: flags 'responsible for', 'helped with', 'worked on'
- Strong verb suggestions: 'Architected', 'Reduced', 'Scaled', 'Delivered'
- Hinglish coaching: 'Yeh bullet weak hai — isko quantify karo!'
- Context-aware: different suggestions for fresher vs senior profiles

4.3 ATS Score Checker

- Analyzes resume against 40+ ATS criteria (keywords, formatting, structure)
- Scores: Overall ATS Score, Keyword Match %, Format Score, Readability
- Identifies: missing sections, problematic formatting, keyword gaps
- Section-level scoring — shows which section is pulling score down
- Actionable fix suggestions for each failing criterion

4.4 JD Keyword Matching & Tailoring

- User pastes or uploads Job Description
- AI extracts must-have and nice-to-have keywords from JD
- Compares against resume — shows matched vs missing keywords
- One-click tailoring: AI rewrites relevant bullets to include missing keywords naturally
- Tailoring score: shows % match improvement after each edit

4.5 AI Bullet Improvement

- Single-click bullet rewrite using STAR format (Situation → Task → Action → Result)
- Adds metrics and quantification where user inputs raw data
- Three rewrite variants shown — user picks best one
- Before/After comparison UI
- Explains WHY the new version is better (educational mode)

4.6 Resume Scoring Engine

- 5-dimension scoring: ATS (30%), Impact (25%), Readability (20%), Keywords (15%), Format (10%)
- Overall score out of 100 with visual gauge
- Percentile comparison: 'Your resume scores better than 67% of resumes for this role'
- Score history tracking — shows improvement over time
- Score breakdown with specific fix recommendations

4.7 Resume Roast Mode

- AI simulates a brutally honest recruiter reviewing the resume
- Gamified tone: direct, funny, but constructive ('Bhai, ye bullet padh ke koi hire nahi karega')
- Section-by-section roast with specific improvement suggestions
- Shareable roast report for social media
- Premium feature — high engagement / virality driver

4.8 AI Recruiter Simulation

- AI simulates a recruiter for the target company/role
- Asks resume-based screening questions
- Evaluates answers and gives feedback
- Highlights resume gaps that a recruiter would catch
- Premium feature — bridges resume builder and interview prep

4.9 Multi-Language Support

- UI available in English and Hindi (Phase 1)
- AI coaching messages in Hinglish for Indian users
- Resume content stays in English (industry standard)
- Error messages, tips, and guidance in preferred language
- Phase 2: Tamil, Telugu, Bengali interface support

4.10 Template System

- 15+ ATS-friendly templates (Phase 1)
- Categories: Fresher, Technical, Creative, Executive, Academic
- All templates verified ATS-parseable (no tables, no text boxes)
- Live template switching — content preserved across template changes
- Color and font customization within ATS-safe parameters

5. AI System Design & Architecture

Micro-AI architecture, prompt engineering, and cost optimization

5.1 Micro-AI Architecture

ResumeAI uses a modular micro-AI architecture where each feature has its own dedicated AI pipeline with specialized prompts, context, and output parsers. This prevents prompt interference and enables independent cost optimization.

AI Module	Function	Model	Avg Cost
BulletImprover	Rewrites weak bullets	gpt-4o-mini	~200 tokens/call
ATSAnalyzer	Scores resume against ATS criteria	gpt-4o	~800 tokens/call
JDExtractor	Extracts keywords from JD	gpt-4o-mini	~400 tokens/call
ResumeCoach	Conversational resume building	gpt-4o	~600 tokens/call
RoastEngine	Brutally honest resume review	gpt-4o	~1000 tokens/call
ScoringEngine	Multi-dimensional resume scoring	gpt-4o	~700 tokens/call
QuestionEngine	Extracts achievements via questions	gpt-4o-mini	~300 tokens/call
RecruiterSim	Simulates recruiter interview	gpt-4o	~500 tokens/call

5.2 Prompt Engineering Strategy

- **System Role Injection:** Each AI module has a specialized system prompt that establishes expert persona (e.g., 'You are a senior tech recruiter at a Fortune 500 company...')
- **Structured JSON Output:** All AI responses are requested as JSON with strict schema — prevents hallucination and enables reliable parsing
- **Few-Shot Examples:** Each prompt includes 2-3 example input/output pairs for consistent formatting
- **Context Passing:** Resume sections passed as structured JSON context, not raw text
- **Temperature Control:** Creative tasks (roast, suggestions): temp 0.7 | Scoring/ATS: temp 0.1 for consistency
- **Token Budget Enforcement:** Max tokens set per module to control costs — 200 for bullets, 1000 for full analysis

5.3 Context Memory Strategy

ResumeAI maintains session-level context across the resume building conversation. A rolling context window stores the last 10 exchanges + current resume state. Redis cache stores session context (TTL: 2 hours). Full context saved to MySQL ai_sessions table for premium users' history feature.

5.4 Hallucination Prevention

- AI never invents job titles, companies, or dates — only rewrites user-provided content
- Strict JSON schema validation on all AI outputs — malformed responses trigger retry
- Achievement metrics only suggested when user provides a base number to quantify
- All AI suggestions shown as 'suggestions' — user must explicitly accept changes

- Confidence scoring: low-confidence outputs flagged with 'Please verify this suggestion'

5.5 Cost Optimization Strategy

Strategy	Implementation
Model Tiering	Use gpt-4o-mini for simple tasks (bullets, Q&A;) — 10x cheaper than gpt-4o
Response Caching	Cache identical ATS analyses (same resume hash) in Redis — 40% cost reduction
Streaming	Stream AI responses to frontend — improves perceived performance
Batch Analysis	Group multiple section analyses into one API call where possible
Free Tier Limits	Free users: 10 AI suggestions/day; Pro: unlimited — controls AI spend
Token Monitoring	Per-user token tracking in MySQL — alert at 80% of plan limit

6. UX Flow & User Journey

Complete interaction flows for every major user path

6.1 Onboarding Flow

- Step 1:** Landing page → 'Build Free Resume' CTA (no signup required for preview)
- Step 2:** Quick profile: Name, Target Role, Experience Level (3 options: Student / 0-2 yrs / 2+ yrs)
- Step 3:** Template selection (3 recommended based on profile)
- Step 4:** Email signup shown AFTER first section is filled (value-first onboarding)
- Step 5:** Google OAuth or Email/Password — JWT token issued
- Step 6:** User lands on Resume Editor with pre-filled name and role

6.2 Resume Creation Flow

- Step 1:** Left: Section navigator (Contact, Summary, Experience, Education, Skills, Projects)
- Step 2:** Center: Live resume preview (real-time rendering)
- Step 3:** Right: AI Coach panel (suggestions, tips, score)
- Step 4:** Each section has AI-guided mode (questionnaire) or manual mode
- Step 5:** AI questionnaire: 'Tell me about your most impactful project' → extracts structured data
- Step 6:** Progress bar shows resume completeness score
- Step 7:** Auto-save every 30 seconds to backend

6.3 AI Coaching Interaction Flow

- Step 1:** User types in experience bullet → 500ms debounce → AI analyzes
- Step 2:** AI suggestion appears as inline ghost text OR right panel card
- Step 3:** User can: Accept (Tab key), Dismiss (Esc), See 3 alternatives, Ask why
- Step 4:** 'Why is this better?' shows educational explanation in Hinglish
- Step 5:** Weak phrases highlighted in yellow underline — click to get improvement
- Step 6:** AI Coach chat at bottom: user can ask 'How do I describe my internship at TCS?'

6.4 JD Upload & Tailoring Flow

- Step 1:** User clicks 'Tailor for Job' → paste JD or upload PDF
- Step 2:** AI extracts: Required Skills, Preferred Skills, Role Keywords, Company Values
- Step 3:** Side-by-side view: Resume keywords vs JD keywords
- Step 4:** Missing keywords shown in red — click each to see suggested placement
- Step 5:** One-click 'Auto-Tailor' rewrites bullets to include top 5 missing keywords
- Step 6:** Before/After comparison with match score improvement shown

6.5 Export Flow

- Step 1:** Click Export → choose: PDF (ATS-safe) / PDF (Visual) / DOCX
- Step 2:** ATS check runs automatically before export
- Step 3:** If ATS score < 60: warning modal with top 3 fixes suggested
- Step 4:** User can export anyway or fix first
- Step 5:** PDF generated server-side (Puppeteer) for pixel-perfect rendering
- Step 6:** Download link valid 24 hours, stored in exports table

7. Database Design (MySQL)

Complete schema with relationships, indexes, and design rationale

Engine: MySQL 8.x — InnoDB engine for all tables. UTF8MB4 charset for emoji/Unicode support. All FKs with ON DELETE CASCADE unless noted.

7.1 Entity Relationship Overview

Core entities: users → resumes → resume_versions → resume_sections. AI layer: ai_feedback → resume_sections. Job matching: job_descriptions ← ats_scores → resumes. Monetization: subscriptions → users, usage_logs → users. Export: exports → resume_versions. Templates: templates (independent).

Table: users

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
email	VARCHAR(255) UNIQUE NOT NULL
password_hash	VARCHAR(255) NULL — bcrypt, NULL for OAuth
full_name	VARCHAR(150) NOT NULL
avatar_url	VARCHAR(500) NULL
target_role	VARCHAR(100) NULL — e.g. 'Software Engineer'
experience_level	ENUM('student','junior','mid','senior') DEFAULT 'student'
preferred_lang	ENUM('en','hi') DEFAULT 'en'
oauth_provider	VARCHAR(30) NULL — 'google' 'github'
oauth_uid	VARCHAR(200) NULL
is_active	TINYINT(1) DEFAULT 1
email_verified	TINYINT(1) DEFAULT 0
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME ON UPDATE CURRENT_TIMESTAMP

Table: resumes

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL — FK users(id)
title	VARCHAR(200) NOT NULL — e.g. 'SWE Resume v2'
template_id	INT UNSIGNED NOT NULL — FK templates(id)
status	ENUM('draft','active','archived') DEFAULT 'draft'
ats_score	TINYINT UNSIGNED NULL — cached latest score (0-100)
overall_score	TINYINT UNSIGNED NULL — cached composite score
is_primary	TINYINT(1) DEFAULT 0 — user's main resume
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME ON UPDATE CURRENT_TIMESTAMP
INDEX	idx_user_id (user_id)
INDEX	idx_status (status)

Table: resume_versions

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
resume_id	INT UNSIGNED NOT NULL — FK resumes(id)
version_num	SMALLINT UNSIGNED NOT NULL — 1, 2, 3...
snapshot	JSON NOT NULL — complete resume data snapshot
label	VARCHAR(100) NULL — e.g. 'Before JD tailoring'
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
INDEX	idx_resume_id (resume_id)
UNIQUE	unique_version (resume_id, version_num)

Table: resume_sections

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
resume_id	INT UNSIGNED NOT NULL — FK resumes(id)
section_type	ENUM('contact','summary','experience','education','skills','projects','certifications','custom')
section_order	TINYINT UNSIGNED DEFAULT 0 — display order
content	JSON NOT NULL — structured section data
is_visible	TINYINT(1) DEFAULT 1
ai_score	TINYINT UNSIGNED NULL — section-level AI score
updated_at	DATETIME ON UPDATE CURRENT_TIMESTAMP
INDEX	idx_resume_section (resume_id, section_type)

Table: ai_feedback

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL — FK users(id)
section_id	INT UNSIGNED NULL — FK resume_sections(id)
feedback_type	ENUM('bullet_improve','ats_check','keyword_match','score','roast','coach_chat','recruiter_sim')
original_text	TEXT NULL
ai_suggestion	TEXT NOT NULL
ai_model	VARCHAR(50) — e.g. 'gpt-4o-mini'
tokens_used	SMALLINT UNSIGNED DEFAULT 0
accepted	TINYINT(1) NULL — NULL=pending, 1=accepted, 0=rejected
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
INDEX	idx_user_feedback (user_id, feedback_type)
INDEX	idx_section_feedback (section_id)

Table: job_descriptions

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL — FK users(id)
title	VARCHAR(200)
company	VARCHAR(150)
raw_text	TEXT NOT NULL
extracted_data	JSON NULL — {required_skills, preferred_skills, keywords, role_level}
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP

Table: ats_scores

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
resume_id	INT UNSIGNED NOT NULL — FK resumes(id)
jd_id	INT UNSIGNED NULL — FK job_descriptions(id), NULL for general ATS check
overall_score	TINYINT UNSIGNED NOT NULL
keyword_score	TINYINT UNSIGNED NOT NULL
format_score	TINYINT UNSIGNED NOT NULL
impact_score	TINYINT UNSIGNED NOT NULL
readability_score	TINYINT UNSIGNED NOT NULL
breakdown	JSON NOT NULL — detailed per-criterion scores and fixes
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
INDEX	idx_resume_ats (resume_id)

Table: templates

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
name	VARCHAR(100) NOT NULL
category	ENUM('fresher','technical','creative','executive','academic')
thumbnail_url	VARCHAR(500)
html_template	TEXT NOT NULL — Handlebars/Mustache template string
css_styles	TEXT NOT NULL
is_premium	TINYINT(1) DEFAULT 0
is_active	TINYINT(1) DEFAULT 1
sort_order	TINYINT UNSIGNED DEFAULT 0

Table: exports

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL — FK users(id)
resume_id	INT UNSIGNED NOT NULL — FK resumes(id)
version_id	INT UNSIGNED NULL — FK resume_versions(id)
export_type	ENUM('pdf_at', 'pdf_visual', 'docx')
file_url	VARCHAR(500) NOT NULL — server path or CDN URL
file_size_kb	SMALLINT UNSIGNED
expires_at	DATETIME — link expiry
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
INDEX	idx_user_exports (user_id)

Table: subscriptions

Column / Constraint	Definition
id	INT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL UNIQUE — FK users(id)
plan	ENUM('free','starter','pro','enterprise') DEFAULT 'free'
razorpay_sub_id	VARCHAR(200) NULL — Razorpay subscription ID
razorpay_customer_id	VARCHAR(200) NULL
status	ENUM('active','past_due','canceled','trialing') DEFAULT 'active'
current_period_start	DATETIME NULL
current_period_end	DATETIME NULL
cancel_at_period_end	TINYINT(1) DEFAULT 0
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME ON UPDATE CURRENT_TIMESTAMP

Table: usage_logs

Column / Constraint	Definition
id	BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY
user_id	INT UNSIGNED NOT NULL — FK users(id)
action	VARCHAR(50) — e.g. 'bullet_improve', 'ats_check', 'export_pdf'
metadata	JSON NULL — {model, tokens, resume_id, jd_id}
created_at	DATETIME DEFAULT CURRENT_TIMESTAMP
INDEX	idx_user_action (user_id, action)
INDEX	idx_created_at (created_at) — for daily analytics

8. API Design

RESTful API endpoints — Node.js / Express backend

Convention: Base URL: /api/v1/ — JSON request/response — JWT Bearer auth — Rate limiting: 60 req/min free, 300 req/min pro

8.1 Authentication APIs

Method	Endpoint	Description
POST	/auth/register	Register user — {email, password, full_name}
POST	/auth/login	Login — {email, password} → {access_token, refresh_token}
POST	/auth/refresh	Refresh JWT — {refresh_token} → new access_token
POST	/auth/logout	Revoke refresh token
GET	/auth/google	Google OAuth redirect
GET	/auth/google/callback	OAuth callback — issues JWT
POST	/auth/forgot-password	Send reset email
POST	/auth/reset-password	Reset with token — {token, new_password}

8.2 Resume CRUD APIs

Method	Endpoint	Description
GET	/resumes	List user's resumes
POST	/resumes	Create resume — {title, template_id}
GET	/resumes/:id	Get full resume with all sections
PUT	/resumes/:id	Update resume metadata
DELETE	/resumes/:id	Soft-delete resume
GET	/resumes/:id/versions	List version history
POST	/resumes/:id/versions	Save snapshot as new version
POST	/resumes/:id/sections	Add/update a section — {section_type, content}
PUT	/resumes/:id/sections/:sid	Update section content
DELETE	/resumes/:id/sections/:sid	Remove section

8.3 AI Feature APIs

Method	Endpoint	Description / Response
POST	/ai/bullet/improve	Rewrite bullet — {text, role, experience_level} → {suggestions[3], explanation}
POST	/ai/resume/score	Score resume — {resume_id} → {overall, ats, impact, readability, keywords, breakdown}
POST	/ai/resume/coach	AI chat — {message, session_id, resume_id} → {reply, suggestions}
POST	/ai/resume/roast	Roast resume — {resume_id} → {roast_sections, overall_verdict}
POST	/ai/resume/question	Get next question — {resume_id, section, history} → {question, purpose}
POST	/ai/recruiter/simulate	Recruiter sim — {resume_id, jd_id, question} → {response, gap_analysis}

8.4 ATS & JD APIs

Method	Endpoint	Description
POST	/ats/analyze	{resume_id, jd_id?} → full ATS report with fix suggestions
POST	/jd/extract	{raw_text} → {required_skills, keywords, role_level, nice_to_have}
POST	/jd/match	{resume_id, jd_id} → {match_score, missing_keywords, matched_keywords}
POST	/jd/tailor	{resume_id, jd_id, auto_apply?} → {tailored_sections, score_improvement}

8.5 Export & Payment APIs

Method	Endpoint	Description
POST	/export/pdf	{resume_id, type: 'ats' 'visual'} → {download_url, expires_at}
POST	/export/docx	{resume_id} → {download_url}
GET	/subscriptions/plans	List available plans
POST	/subscriptions/create	{plan_id} → Razorpay checkout session
POST	/webhooks/razorpay	Razorpay payment webhook handler
GET	/subscriptions/me	Current user subscription status

9. AI Prompt Engineering System

Production-grade prompts for every AI module

9.1 Bullet Improvement Prompt

Role: SYSTEM

Prompt Template:

```
You are a senior technical recruiter and resume coach with 15 years of experience.

Your task: rewrite the given resume bullet point to be stronger, more impactful, and ATS-friendly.

Rules:

1. Start with a strong action verb
2. Add quantifiable metrics where user provides numbers
3. Follow STAR format implicitly (Action → Result)
4. Never invent numbers or achievements
5. Keep it under 20 words

Output JSON: {"suggestions": [str, str, str], "explanation": str, "weak_phrases_found": [str]}

User context: Role={role}, Experience={level}
```

9.2 ATS Analysis Prompt

Role: SYSTEM

Prompt Template:

```
You are an ATS (Applicant Tracking System) expert.

Analyze the provided resume JSON and score it on these criteria:

- keyword_score (0-100): presence of role-relevant keywords
- format_score (0-100): ATS-parseable structure, no tables/graphics
- contact_score (0-100): email, phone, LinkedIn present
- experience_score (0-100): dates, titles, companies structured correctly
- skills_score (0-100): skills section present and well-formatted

Output JSON: {"overall": int, "scores": {}, "critical_issues": [str], "fixes": [str]}

Be deterministic. Temperature: 0.1
```

9.3 Resume Roast Prompt

Role: SYSTEM

Prompt Template:

```
You are a brutally honest but constructive resume reviewer.

Your style: Direct, occasionally humorous (like a friendly senior who wants you to succeed).

For Indian audience: Hinglish is encouraged in roast messages.

For each section, give: rating (1-5), roast (1-2 sentences), fix (specific improvement).

End with: overall_verdict and top_3_fixes.

Output JSON: {"sections": [{name, rating, roast, fix}], "overall_verdict": str, "top_3_fixes": [str]}
```

Example roast tone: "Yeh summary padh ke lagta hai ChatGPT ne 10 second mein likha."

9.4 Prompt Version Control Strategy

- All prompts stored in /src/ai/prompts/ as .ts files with version comments
- Prompt changes go through A/B testing — 50% traffic on new prompt, compare acceptance rates
- Prompt performance logged: tokens used, user acceptance rate, latency
- Rollback mechanism: feature flag switches between prompt versions

10. Resume Scoring Engine

Multi-dimensional weighted scoring system

10.1 Scoring Dimensions & Weights

Dimension	Weight	What We Measure
ATS Compatibility	30%	Keyword presence, format parsability, section structure, file compatibility
Impact Score	25%	Action verbs, quantified achievements, STAR format adherence
Keyword Relevance	20%	Role-specific keywords, industry terms, skill match to target role
Readability	15%	Sentence length, clarity, grammar quality, professional tone
Format & Structure	10%	Visual hierarchy, section completeness, appropriate length

10.2 ATS Scoring Criteria (30 points breakdown)

Criterion	Max Points	Detection Method
Email present	3pts	Simple regex check
Phone present	2pts	Indian format: +91XXXXXXXXXX
LinkedIn URL	2pts	linkedin.com/in/ pattern match
No tables/columns	4pts	HTML/JSON structure analysis
Standard section names	4pts	Experience, Education, Skills — not fancy aliases
Date format consistent	3pts	Month YYYY or YYYY format
No images/graphics	3pts	Template analysis
File parsability	4pts	PDF text layer present
Skills section	3pts	Dedicated skills section present
Action verbs used	2pts	Starts bullets with verb from action verb list

10.3 Impact Scoring Rubric

Factor	Points	Detection
Quantified achievement	+3pts each	Numbers/% in bullet
Strong action verb	+2pts each	From curated list of 150 power verbs
STAR structure	+2pts	Implied action + result in bullet
Passive voice	-2pts each	'Was responsible for', 'Assisted with'
Weak filler phrases	-1pt each	'Various', 'several', 'helped with'

Factor	Points	Detection
Responsibilities-only	-5pts total	No achievement mentioned across experience

10.4 Score Interpretation

Score	Grade	Recommendation
85–100	Excellent	Ready to apply — strong for ATS and recruiter review
70–84	Good	Minor improvements recommended — apply to most roles
55–69	Fair	Several issues to fix before applying — use AI suggestions
40–54	Weak	Significant rework needed — use Resume Coach flow
0–39	Poor	Start over with AI-guided building — multiple critical issues

11. Security & Privacy

11.1 Authentication & Authorization

- **JWT Access Tokens:** 15-minute expiry — short-lived to limit exposure
- **Refresh Tokens:** 7-day expiry, stored httpOnly cookie — not accessible to JS
- **Password Hashing:** bcrypt with cost factor 12
- **OAuth Security:** PKCE flow for Google OAuth — prevents auth code interception
- **Rate Limiting:** express-rate-limit: 5 login attempts / 15 min per IP — blocks brute force
- **Role Guards:** Middleware validates subscription tier before AI feature access

11.2 Data Security

- All API communication over HTTPS (TLS 1.3)
- Resume content stored as JSON in MySQL — no raw file storage
- Uploaded JD files scanned for malware before processing — multer + file-type validation
- User photos / avatars stored with randomized filenames — no enumerable URLs
- SQL injection prevention: Sequelize ORM with parameterized queries throughout
- XSS prevention: helmet.js headers, DOMPurify on all user-rendered content

11.3 AI Privacy

- Resume data sent to OpenAI API — only the specific section being analyzed, never full PII
- No user data used for OpenAI model training (Enterprise API agreement)
- User can request 'AI amnesia' — clears all stored AI feedback from DB
- AI suggestions are ephemeral by default — only stored if user accepts

11.4 Privacy Policy & Compliance

Area	Implementation
Data Deletion	Full account + resume deletion within 24 hours of request
Data Export	User can export all their data as JSON (GDPR Art. 20)
Consent	Explicit consent before AI processing of resume data
Cookies	Only essential + analytics cookies; user controls in settings
Minors	No users under 16; age confirmation at registration
Indian IT Act	Compliant with IT Act 2000 and DPDP Act 2023 requirements

12. Monetization Strategy

India-first freemium model with Razorpay integration

12.1 Pricing Tiers

Feature	Free ₹0/month	Starter ₹199/month	Pro ₹499/month	Enterprise Custom
Resumes	1	3	Unlimited	Unlimited
Templates	3 basic	8	All 15+	Custom
AI Suggestions/day	10	50	Unlimited	Unlimited
ATS Score Checks	2/month	15/month	Unlimited	Unlimited
JD Tailoring	✗	5/month	Unlimited	Unlimited
Resume Roast	✗	1/month	Unlimited	Unlimited
AI Recruiter Sim	✗	✗	✓	✓
Version History	✗	5 versions	Unlimited	Unlimited
PDF Export	1 (watermark)	Unlimited	Unlimited	Unlimited
Priority Support	✗	Email	Chat + Email	Dedicated
College Dashboard	✗	✗	✗	✓

12.2 Revenue Streams

- **Subscription (Primary):** MRR from Starter + Pro plans — target 5,000 paying users in Year 1
- **College Partnerships:** ₹50,000–₹2L/year per institution for placement cell dashboards
- **Affiliate Revenue:** Job portal referrals (Naukri, LinkedIn) — ₹50–200 per successful referral
- **Resume Writing Service:** AI-assisted human review at ₹999 — hybrid upsell
- **API Licensing:** Phase 3 — license ATS scoring API to bootcamps and coaching institutes

12.3 Conversion Funnel Strategy

- Value-first onboarding: resume preview before signup — reduces friction
- AI suggestion limit hit: contextual upsell ('Upgrade to get 40 more suggestions today')
- ATS score shown partially free, full breakdown is Pro — creates curiosity gap
- Resume Roast is viral feature — shareable social content drives organic acquisition
- Annual plan: 2 months free (₹1,990/yr Starter, ₹4,990/yr Pro) — improves LTV

13. MVP Roadmap — Phase 1 → 3

Phase 1 — Foundation MVP (Weeks 1–8)

Feature	Priority	Description
Auth System	Must	JWT + Google OAuth, email verification
Resume Builder	Must	Section-based editor, 5 core sections
3 Templates	Must	ATS-safe Fresher, Technical, Clean templates
AI Bullet Improvement	Must	Single bullet rewrite, 3 suggestions
Basic ATS Score	Must	Rule-based scoring (no AI) — 15 criteria
PDF Export	Must	Puppeteer-based, ATS-safe PDF
Free Tier	Must	1 resume, 10 AI suggestions/day
MySQL Schema	Must	All core tables from Section 7

Phase 2 — AI Core (Weeks 9–16)

Feature	Priority	Description
Full AI Scoring	Must	GPT-powered multi-dimension scoring
JD Upload + Matching	Must	Keyword extraction + match score
JD Tailoring	Must	AI rewrites bullets for JD alignment
Resume Coach Chat	Must	Conversational AI in editor sidebar
AI Questioning System	Must	Interview-style achievement extraction
Version History	Must	Snapshot + restore
Razorpay Payments	Must	Starter + Pro subscription billing
Resume Roast Mode	Should	Viral premium feature
10+ Templates	Should	Expand template library
Hinglish UI mode	Should	Toggle EN/HI interface

Phase 3 — Growth (Weeks 17–28)

Feature	Priority	Description
AI Recruiter Simulation	Must	Premium feature for interview prep bridge
Analytics Dashboard	Must	User score trends, AI usage, export history
College B2B Portal	Should	Placement cell management dashboard

Feature	Priority	Description
LinkedIn Optimization	Should	AI headline + about section rewriter
Mobile PWA	Should	Offline-capable progressive web app
Team Collaboration	Could	Share resume with mentor for review
API Public Beta	Could	ATS scoring API for third parties
Interview Prep Module	Future	AI-generated questions from resume

14. Technical Architecture

Production-grade monorepo architecture

14.1 Monorepo Structure

Project Structure:

```
resume-ai/
├── apps/
│   ├── web/ # Next.js 14 frontend (App Router)
│   └── api/ # Node.js + Express backend
├── packages/
│   ├── ai-engine/ # AI service layer (shared)
│   ├── pdf-service/ # Puppeteer PDF generation
│   ├── scoring-engine/ # Resume scoring logic
│   └── shared-types/ # TypeScript interfaces
├── docker-compose.yml
└── turbo.json # Turborepo build config
```

14.2 Frontend Architecture (Next.js 14)

Concern	Solution
App Router	app/ directory — server components for SEO pages, client for editor
State Mgmt	Zustand — resume editor state, AI suggestions queue, user session
Rich Text	TipTap editor — ProseMirror-based, extensible for AI inline suggestions
PDF Preview	react-pdf — live resume preview in editor
AI Streaming	Vercel AI SDK — streams GPT responses to coach chat UI
Styling	Tailwind CSS + shadcn/ui component library
Auth	next-auth with JWT strategy — cookies for session

14.3 Backend Architecture (Node.js + Express)

Concern	Solution
API Layer	Express.js — RESTful, versioned (/api/v1/)
ORM	Sequelize + MySQL2 — type-safe queries, migrations
Auth	jsonwebtoken + passport-jwt + passport-google-oauth20
AI Service	OpenAI Node SDK — modular service classes per AI feature
PDF Gen	Puppeteer — headless Chrome for pixel-perfect PDF rendering
Queue	Bull + Redis — PDF generation and email jobs queued async

Concern	Solution
Validation	Zod — runtime schema validation on all API inputs
Logging	Winston + Morgan — structured JSON logs

14.4 Caching Strategy

- **Redis Layer 1:** AI response cache — hash(resume_section + prompt_version) → cache 1 hour
- **Redis Layer 2:** ATS score cache — hash(resume_content) → cache 30 min
- **Redis Layer 3:** Session data + JWT denylist for logout
- **MySQL Query Cache:** Frequently read templates and subscription plans cached at app start

14.5 PDF Generation Pipeline

1. Export request → Bull queue job created → 2. Worker picks up job → 3. Puppeteer renders resume HTML template with user data → 4. PDF saved to /uploads/exports/{user_id}/{uuid}.pdf → 5. Export record created in DB with expiry → 6. Download URL returned to frontend via WebSocket notification

14.6 Environment Variables

.env structure:

```
# Database
DB_HOST=localhost
DB_PORT=3306
DB_NAME=resumeai
DB_USER=resumeai_user
DB_PASSWORD=

# Auth
JWT_ACCESS_SECRET=<256-bit-secret>
JWT_REFRESH_SECRET=
JWT_ACCESS_EXPIRY=15m
JWT_REFRESH_EXPIRY=7d
GOOGLE_CLIENT_ID=
GOOGLE_CLIENT_SECRET=

# AI
OPENAI_API_KEY=sk-...
OPENAI_ORG_ID=org-...
AI_DAILY_BUDGET_USD=50

# Payments
RAZORPAY_KEY_ID=rzp_live...
RAZORPAY_KEY_SECRET=

# Redis
REDIS_URL=redis://localhost:6379

# App
```

```
NEXT_PUBLIC_APP_URL=https://resumeai.in
```

```
NODE_ENV=production
```


15. Analytics & Tracking

15.1 Key Metrics to Track

Metric	Definition & Target
Activation Rate	% users who complete first resume section after signup — target 60%
Resume Completion Rate	% resumes with 4+ sections filled — target 45%
AI Acceptance Rate	% AI suggestions accepted by users — target 35%+ means prompts are working
ATS Check Conversion	% users who run ATS check after building — target 70%
Export Conversion	% sessions ending in PDF export — target 40%
Free to Paid	% free users who upgrade within 30 days — target 8%
AI Cost per User	OpenAI spend per MAU — budget █15/user/month
D7 / D30 Retention	Users returning after 7/30 days — target 30% D7, 15% D30

15.2 Analytics Tools Stack

- **Posthog (self-hosted):** Event tracking, session recording, funnel analysis — privacy-compliant
- **Custom MySQL dashboard:** AI usage, token spend, export counts via usage_logs table queries
- **Google Analytics 4:** Traffic sources, device breakdown, geographic data
- **Sentry:** Frontend error tracking + performance monitoring
- **Uptime monitoring:** Better Uptime — API health checks every 30 seconds

16. Risks & Mitigation

Risk	Severity	Scenario	Mitigation
AI Hallucination	High	AI invents job titles or skills	Strict JSON schema + user must accept changes + no auto-apply mode
API Cost Explosion	High	OpenAI costs spike with scale	Per-user daily limits, gpt-4o-mini for simple tasks, aggressive caching
ATS Inconsistency	Medium	Different ATS systems have different rules	Score against common subset (Workday, Greenhouse, Taleo) + disclaimer shown
Generic AI Outputs	Medium	All users get similar-sounding resumes	Few-shot examples with diverse styles, user can request 'different tone'
User Trust	Medium	Users worried AI will 'fake' their resume	Clear messaging: AI improves HOW you say it, never WHAT you did
Content Plagiarism	Low	AI might generate plagiarized phrases	Originality is built into prompts — STAR format is always personalized to input
Razorpay Failures	Low	Payment gateway downtime	Grace period: 3 days before subscription downgrade on payment failure

17. Future Vision & Roadmap

Where ResumeAI goes in Year 2–3

Vision 2027: ResumeAI becomes India's #1 AI career platform — not just resumes, but LinkedIn, interviews, job matching, and career coaching in one intelligent system.

17.1 Product Expansion Roadmap

Feature	Target	Description
LinkedIn Optimizer	Q1 2026	AI rewrites LinkedIn headline, About, and experience sections for searchability
Interview Prep AI	Q2 2026	Resume-based question bank + AI mock interviewer with feedback
Job Matching Engine	Q3 2026	Match user's resume against live Naukri/LinkedIn job listings
AI Career Coach	Q4 2026	Long-term career advisor — skill gap analysis, learning path, salary negotiation
AI Portfolio Builder	Q1 2027	Auto-generated personal website from resume data
Video Resume	Q2 2027	AI-scripted 60-second video resume with teleprompter
Referral Network	Q3 2027	Connect job seekers with alumni at target companies
Salary Intelligence	Q4 2027	AI salary negotiation coach based on role, city, and experience data

17.2 Platform Evolution

- **B2C → B2B2C:** Partner with 500+ colleges for placement portal integration — captive user acquisition
- **Vernacular AI:** Full Hindi, Tamil, Telugu resume coaching — massive untapped Indian market
- **WhatsApp Bot:** Resume review and ATS check via WhatsApp — meet users where they are
- **AI Model Ownership:** Fine-tune open-source LLM on Indian resume corpus — reduce OpenAI dependency
- **Data Moat:** Anonymized resume + outcome data (did they get hired?) → unique training dataset